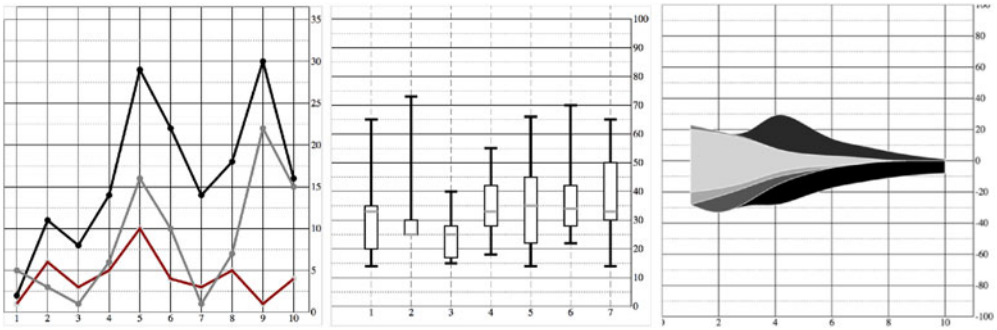


Figure 4.1 The charts we'll create in this chapter using D3 generators and components. From left to right: a line chart, a boxplot, and a streamgraph.

generators. A D3 component, like an axis, is a function for drawing all the graphical elements necessary for an axis. A generator, like `d3.svg.line()`, lets you draw a straight or curved line across many points. The chapter begins by showing you how to add axes to scatterplots as well as create line charts, but before the end you'll create an exotic yet simple chart: the streamgraph. By understanding how D3 generators and components work, you'll be able do more than re-create the charts that other people have made and posted online (many of which they're just re-creating from somewhere else).

A chart (and notice here that I don't use the term *graph* because that's a synonym for *network*) refers to any flat layout of data in a graphical manner. The datapoints, which can be individual values or objects in arrays, may contain categorical, quantitative, topological, or unstructured data. In this chapter we'll use several datasets to create the charts shown in figure 4.1. Although it may seem more useful to use a single dataset for the various charts, as the old saying goes, "Horses for courses," which is to say that different charts are more suitable to different kinds of datasets, as you'll see in this chapter.

4.1 General charting principles

All charts consist of several graphical elements that are drawn or derived from the dataset being represented. These graphical elements may be graphical primitives, like circles or rectangles, or more-complex, multipart, graphical objects like the boxplots we'll look at later in the chapter. Or they may be supplemental pieces like axes and labels. Although you use the same general processes you explored in previous chapters to create any of these elements in D3, it's important to differentiate between the methods available in D3 to create graphics for charts.

You've learned how to directly create simple and complex elements with data-binding. You've also learned how to measure your data and transform it for display. Along with these two types of functions, D3 functionality can be placed into three broader categories: generators, components, and layouts, which are shown in figure 4.2 along with a general overview of how they're used.